FinSight AI

NBFC PLATFORM · FINAL PROJECT PRESENTATION

Every loan file is a stack of paper someone has to read, check and judge.

FinSight AI — agentic loan underwriting for Indian NBFCs

29 CrewAI agents read the documents, check them for fraud and RBI policy, score the risk and draft the credit memo — on a local model, with a person making every decision.

29	12	25 / 29	130
agents in 6 crews	loan products	targets met on real data	automated tests

Shreyas Bile · 4 October 2026

Photo: Dimitri Karastelev / Unsplash

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00

The product

What FinSight does, who uses it, and what it looks like.



IN THIS SECTION

- The problem
- Three roles
- Twelve loan products
- App tour

THE PRODUCT · THE PROBLEM

A loan desk that still runs on paper

Every NBFC loan file is a stack of Indian documents that someone has to read, verify and judge.

Manual document reading
Officers retype PAN, Aadhaar, payslip and statement fields by hand.

Forgeries are hard to see
An edited payslip or a pasted ID photo looks genuine to the eye.

Slow, uneven turnaround
In the seeded process data a file waits 77 h on average before its documents are read.

Rules keep changing
RBI Digital Lending Directions 2025, KFS and penal-charge rules apply to every loan.

What it costs: slow decisions, uneven checks, and forgeries that get through.



Photo: Wesley Tingey / Unsplash

THE PRODUCT

Agents do the reading and checking. People decide.

The same loan file, handled by 29 specialised agents that produce evidence and drafts — never a decision.

Documents uploaded
PAN · Aadhaar · payslip · statement

→

29 agents read and check
one job, one set of tools each

→

Evidence, scores, drafts
fraud tier · policy check · memo · offer

→

A person decides
officer · Head above ₹5 lakh

Reads every document
Fields extracted, formats and checksums validated.

Checks for tampering
Ten signals, incl. a detector trained on real forgeries.

Matches the person
Faces across documents, plus a live selfie.

Fuses the fraud evidence
One advisory score and tier for the case.

Tests RBI and internal rules
Every rule with its citation; missing evidence is 'need info'.

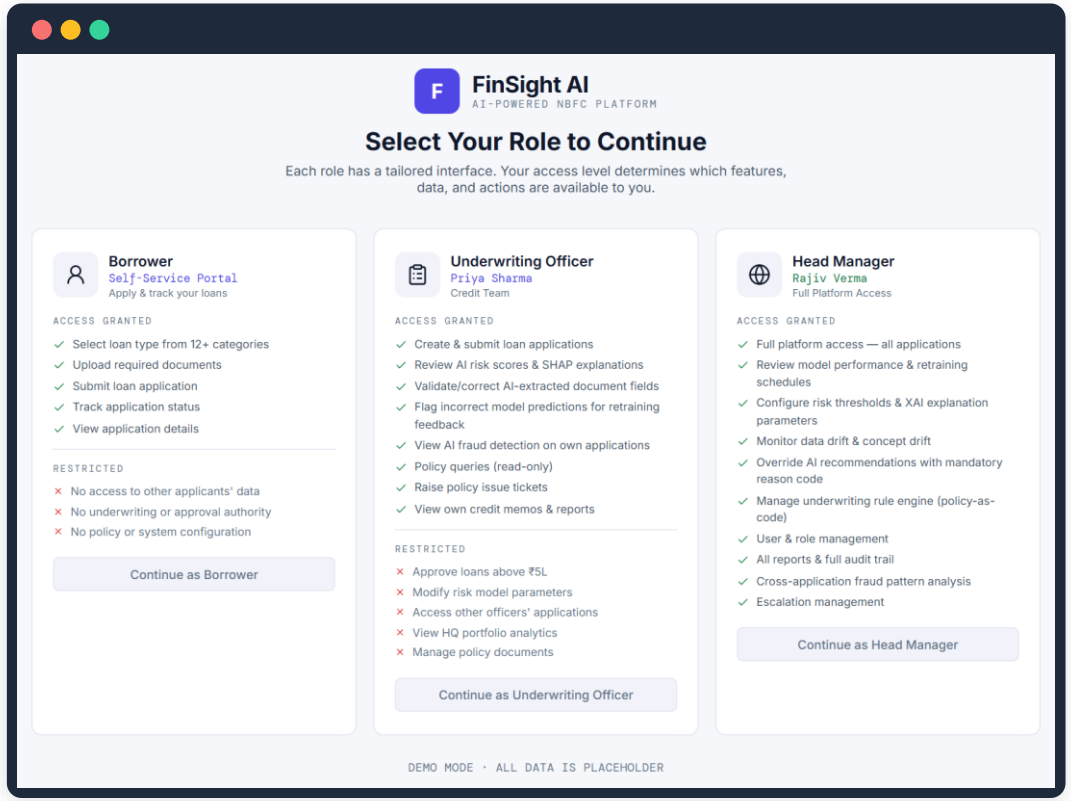
Scores risk, drafts the memo
Risk grade, credit memo, offer and the next step.

People decide. Nothing is auto-rejected, the fraud tier is advice to a reviewer, and any loan above ₹5 lakh needs the Head Manager's approval.

THE PRODUCT

Three roles, twelve loan products

One app, three views. Each product has its own document checklist and its own lifecycle branch.



Borrower

Applies, uploads documents, confirms what the agents read, takes a selfie and submits.

Underwriting Officer

Works the queue: evidence, fraud report, policy check, credit memo — then recommends.

Head Manager

Approves loans above ₹5 lakh, runs fraud triage, manages AI models and RBI rules.

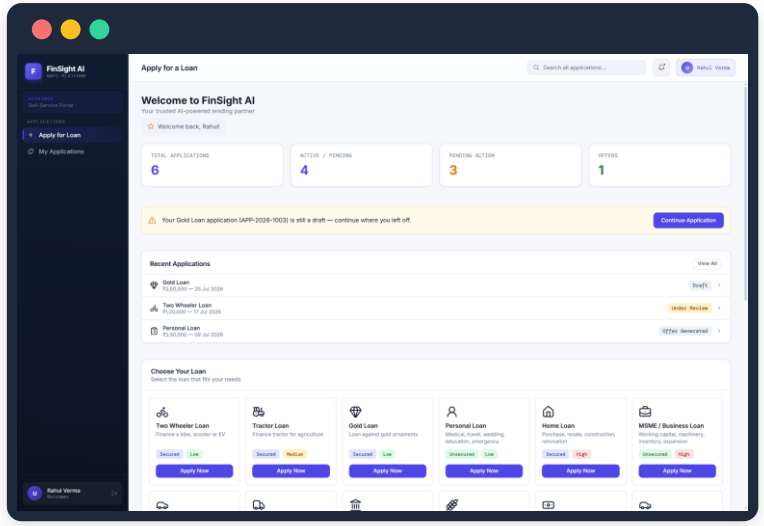
12 LOAN PRODUCTS

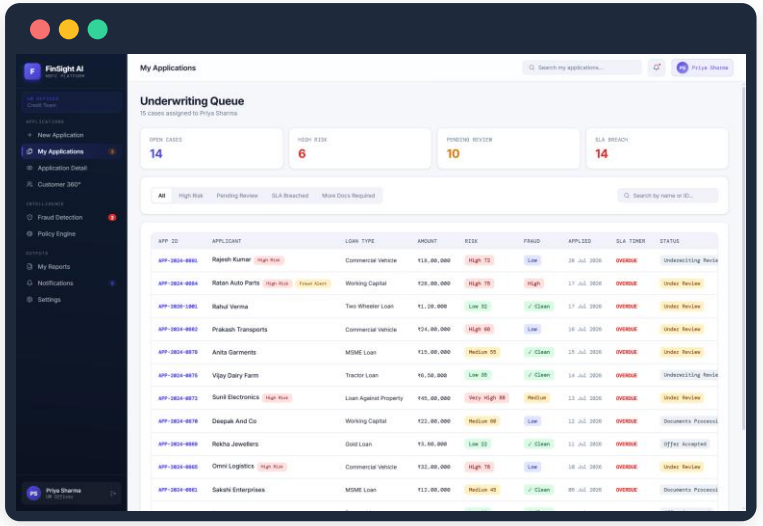
- Two Wheeler
- Tractor
- Gold
- Personal
- Home
- MSME / Business
- Car
- Commercial Vehicle
- Loan Against Property
- Microfinance / Rural
- Working Capital
- Used Vehicle

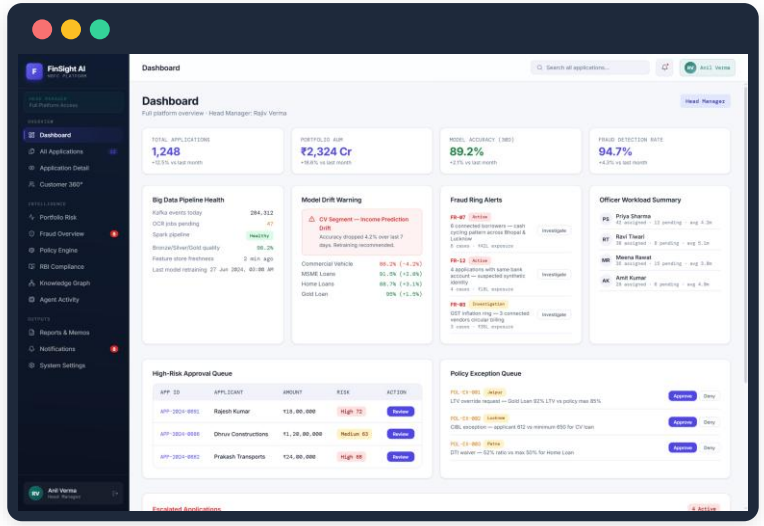
THE PRODUCT

A quick tour of the app

Screens captured from the running app (the application data shown is seeded demo data).







Borrower portal

Status of every application and the 12 products to apply for.

- Apply for any of the 12 products
- Upload, confirm fields, take a selfie
- Track status and offers

Officer queue

Assigned cases with risk, fraud, SLA timer and next action.

- Risk and fraud column for every case
- SLA timer and the next action
- Open a case: 8 tabs of evidence

Head dashboard

Portfolio view: approvals waiting, fraud alerts, model health.

- Approvals waiting above ₹5 lakh
- Fraud alerts and ring triage
- AI model and RBI rule control

01

Architecture

Six layers, one gateway to the language model, and one rule for every agent: tools decide, the model only explains.

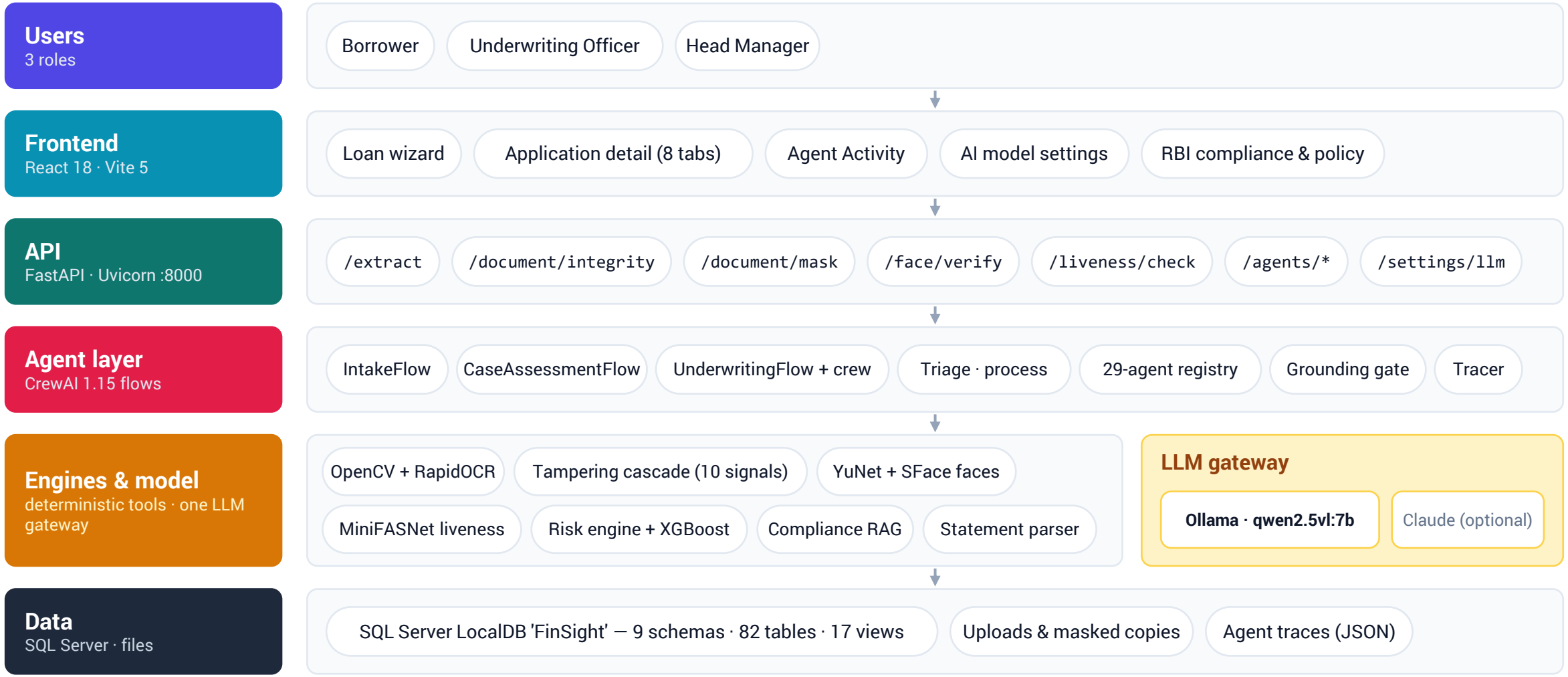


IN THIS SECTION

System layers | Tech stack | How one agent runs

ARCHITECTURE

System architecture



ARCHITECTURE

Tech stack

Everything runs on one laptop — the local model uses an RTX 4060 (8 GB).

Frontend
React 18 · Vite 5 · lucide icons
Role-based single-page app

API
FastAPI 0.115 · Uvicorn · Pydantic 2
Python 3.13

Agents
CrewAI 1.15.22
Flows, crews and 29 registered agents

Language models
Ollama · qwen2.5vl:7b (default)
qwen3-vl:4b option · Claude optional

Vision & OCR
OpenCV 4.10 · RapidOCR (ONNX)
PDFium (pypdfium2) for PDFs

Face & liveness
YuNet · SFace · MiniFASNet
ONNX Runtime 1.27, all local

ML & scoring
XGBoost 3.3 · scikit-learn · PyTorch
Two-wheeler default model · patch forgery detector

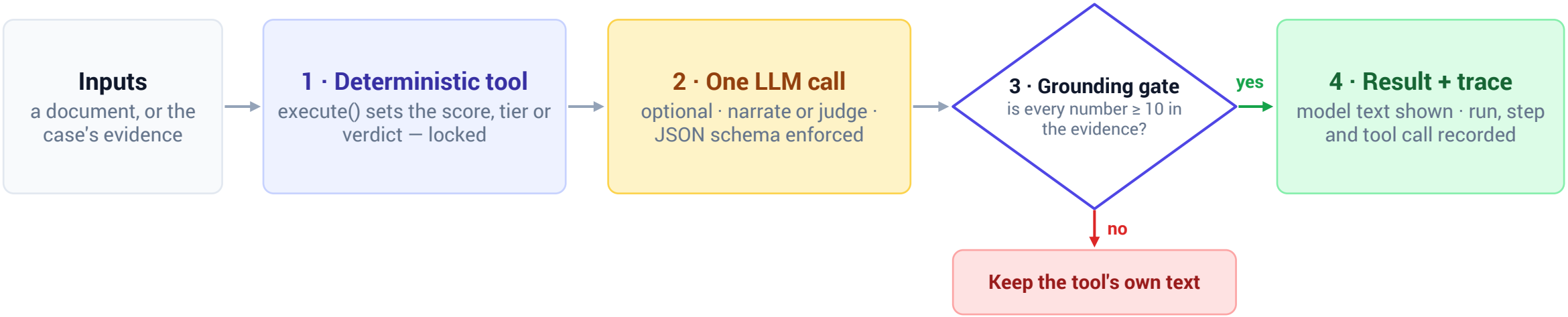
Data
SQL Server LocalDB · pyodbc
9 schemas · 82 tables · 17 views

Quality
pytest · two evaluation harnesses
130 tests · all 29 agents scored on real data

ARCHITECTURE

How one agent runs: tools decide, the model only explains

Every agent follows the same four steps. A score is fixed before any language model is involved.



THREE KINDS OF AGENT

18 Deterministic

Tool only — no model at all.

Validation, masking, face match, tampering signals, fraud fusion, policy rules, risk, offer…

7 Hybrid

Tool decides, model narrates or judges within limits.

Authenticity, tamper explainer, fraud patterns, triage, rule authoring, income, process mining

4 LLM

The model does the core task inside a fenced call.

Extraction, regulatory Q&A, credit memo, underwriting supervisor

02

The 29 agents

Six crews. Each agent has one job, one set of tools and a pass mark fixed before it was tested. Scores shown are from real and public data.



IN THIS SECTION

Agent map | Intake | Privacy & identity | Tampering | Fraud fusion | Patterns & triage | Compliance | Underwriting | Risk | Operations

THE 29 AGENTS

The agent map: 29 agents in six crews

Intake
4 agents

- #1 deterministic — Preprocessing
- #2 LLM — Extraction
- #3 deterministic — Validation
- #4 deterministic — Document Identity

Privacy
1 agent

- #5 deterministic — PII Masking

Identity & fraud evidence
11 agents

- #6 deterministic — Face Verification
- #7 deterministic — Tampering Forensics
- #8 hybrid — Authenticity
- #9 hybrid — Tamper Explainer
- #10 deterministic — Tamper Locator
- #11 deterministic — Cross-Doc Verification
- #12 deterministic — Dedupe
- #13 deterministic — Fraud Fusion
- #14 hybrid — Fraud Patterns
- #15 hybrid — Fraud Triage
- #29 deterministic — Liveness

Compliance
4 agents

- #16 deterministic — KB Ingestion
- #17 hybrid — Rule Authoring
- #18 LLM — Regulatory Q&A
- #19 deterministic — Policy Check

Underwriting
6 agents

- #20 hybrid — Income Analysis
- #21 deterministic — Risk Scoring
- #22 LLM — Credit Memo
- #23 LLM — Underwriting Supervisor
- #24 deterministic — Offer Generation
- #25 deterministic — Workflow Router

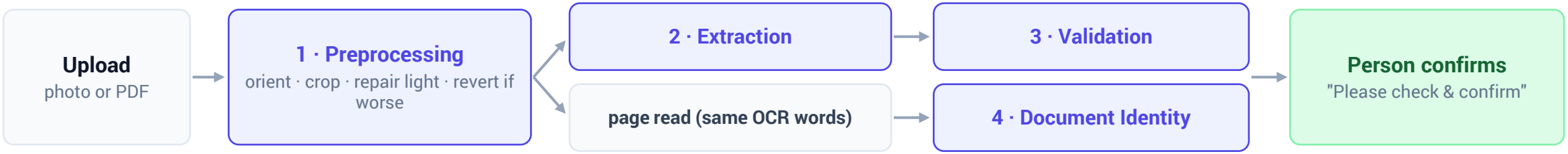
Operations
3 agents

- #26 deterministic — Notifier
- #27 deterministic — Audit / Storage
- #28 hybrid — Process-Mining Analyst

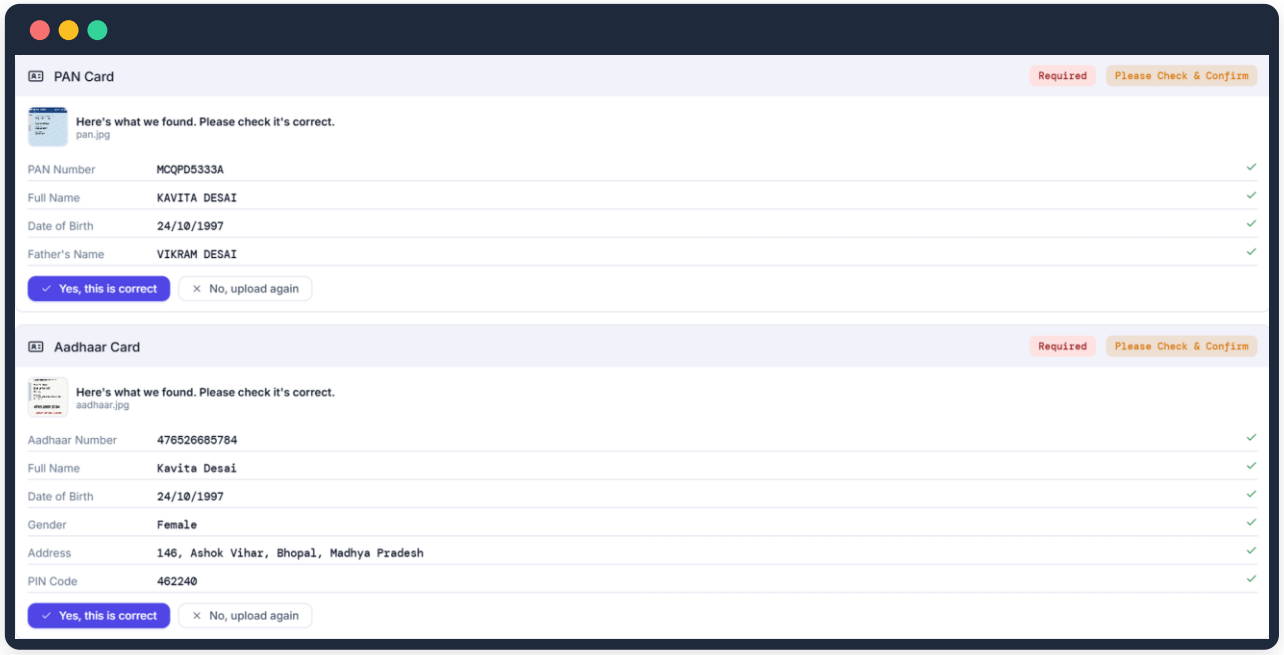
THE 29 AGENTS · INTAKE CREW

Intake: every upload becomes checked, structured fields

Four agents run on each document — about 7 seconds for a PAN card on the local model.



qwen2.5vl reads every requested field into JSON, inventing nothing · Verhoeff, PAN, IFSC, GSTIN formats · the page's own phrases must match the requested type



94.4% — **Extraction**
printed fields correct · 150 PAN cards, 60 statements

94.4% — **Document Identity**
wrong uploads caught · right ones refused 1.6%

100% — **Validation**
183,214 real IFSC codes accepted, mistakes rejected

69.1% — **Preprocessing**
card outline found in phone photos · target 80% (Weak)

Privacy & identity: mask the numbers, match the faces

PII Masking protects the officer's view; Face Verification and Liveness make sure one real person is applying.

PII Masking #5

PAN

~~MCQPD5333A~~ → XXXXXX333A

AADHAAR

XXXX XXXX 5784

- Finds the ID number, even inside a line of text, and masks all but the last four
- Reads the card upright first, so sideways photos can't leak
- A card whose number can't be located is withheld, never shown unmasked

98.4% no full ID readable on 187 of 190 documents · target 100%

Face Verification #6



- YuNet finds each face, SFace turns it into an embedding
- Cosine similarity ≥ 0.363 → same person (0.3% false matches)
- A blurry or tiny face can only be 'uncertain', never 'mismatch'

95.9% correct verdicts · 4 public benchmarks, also printed on ID cards

Liveness #29

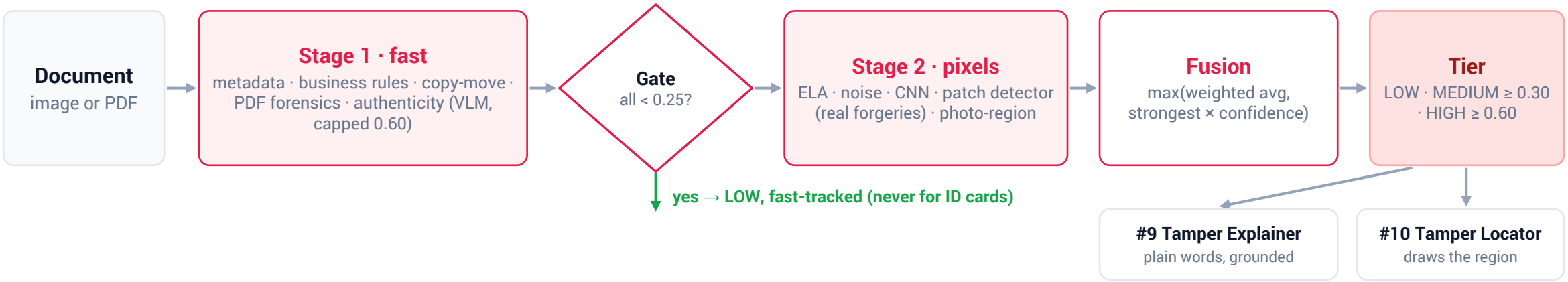


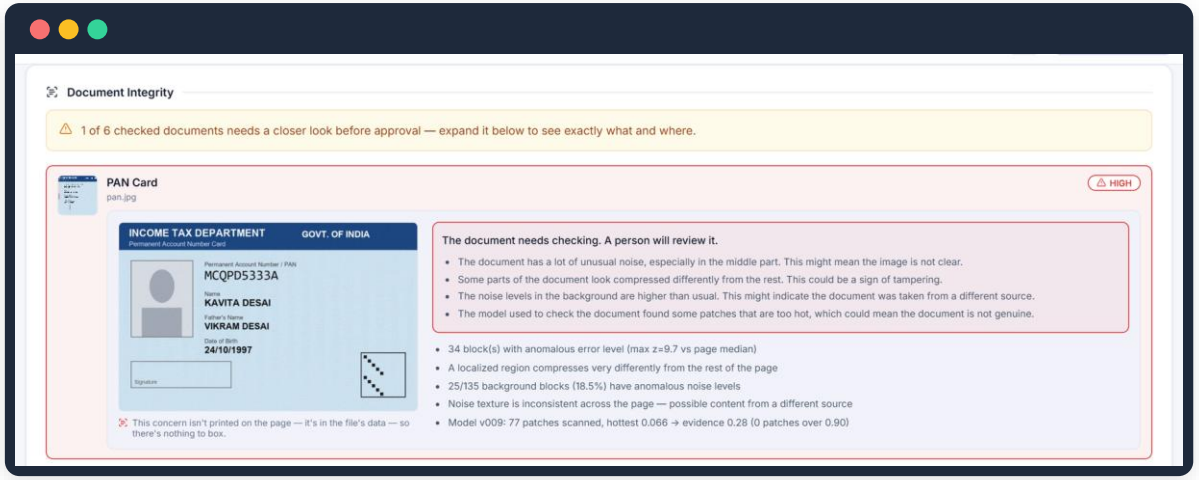
- Selfie checked by MiniFASNet for printed-photo attacks
- Judges the whole frame — a tight face crop loses what it looks at
- The picture is checked and discarded

0.4% ACER on 2,000 webcam frames (target ≤ 10%) · full frames only

Tampering: a ten-signal cascade, fused into one honest tier

Cheap checks first; pixel forensics only when needed (ID cards always get them). Two helper agents explain and locate.





How a flagged document is shown to the officer (synthetic card, captured before the real-data calibration).

88.9% — **Forged documents flagged**
real forged receipts + ID images · was 17%

0.8% — **Genuine rated HIGH**
genuine ID phone photos flagged at all: 0.5%

74.6% — **Locator · Explainer**
forged region boxed · explanations 98.3% grounded

Authenticity (#8, the vision model asked 'was this edited?') flags 11% — Weak: a 7B model cannot see a local edit.

Case assessment: all the evidence, one advisory route

Runs when the file is complete — no language model at all, about 100 ms.

Evidence bundle
extracted fields · tampering tiers · face result · statement rows

#11 Cross-Doc
names · DOB · gender

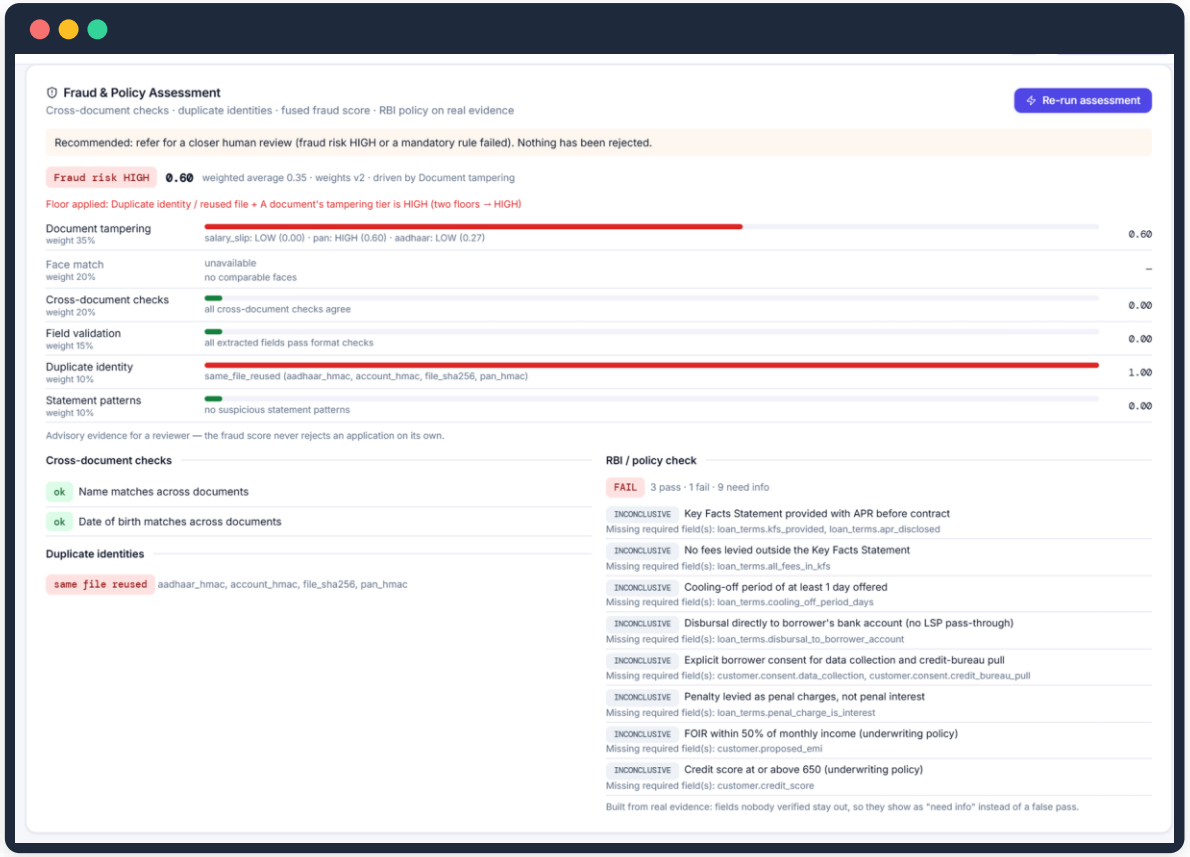
#12 Dedupe
HMAC keys · file hash · face

#14 Patterns
statement rows

#13 Fraud Fusion
one advisory score and tier (LOW · MEDIUM · HIGH)

#19 Policy Check
15 active RBI / internal rules → PASS · FAIL · NEED INFO

Advisory route
proceed · or refer for a closer human review — never reject

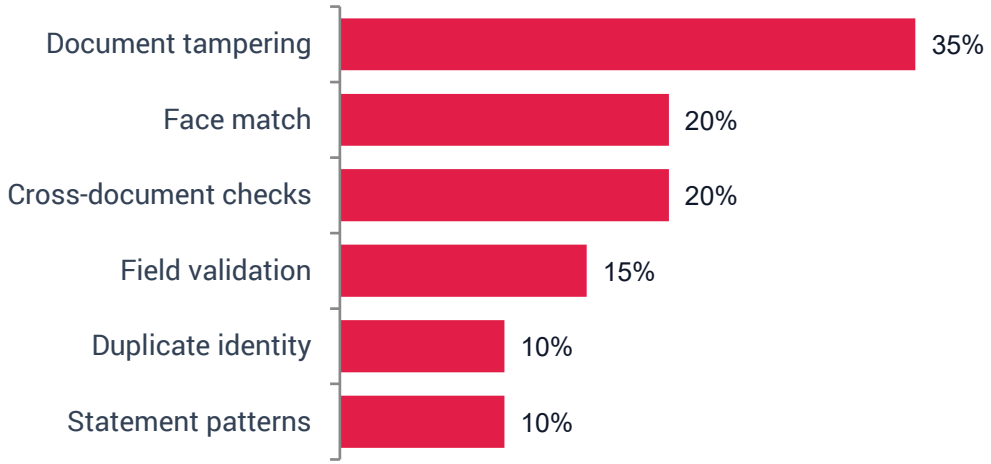


Real run, synthetic applicant submitted twice: Dedupe finds the same files → two floors → HIGH.

Fraud Fusion v2: how one score is built

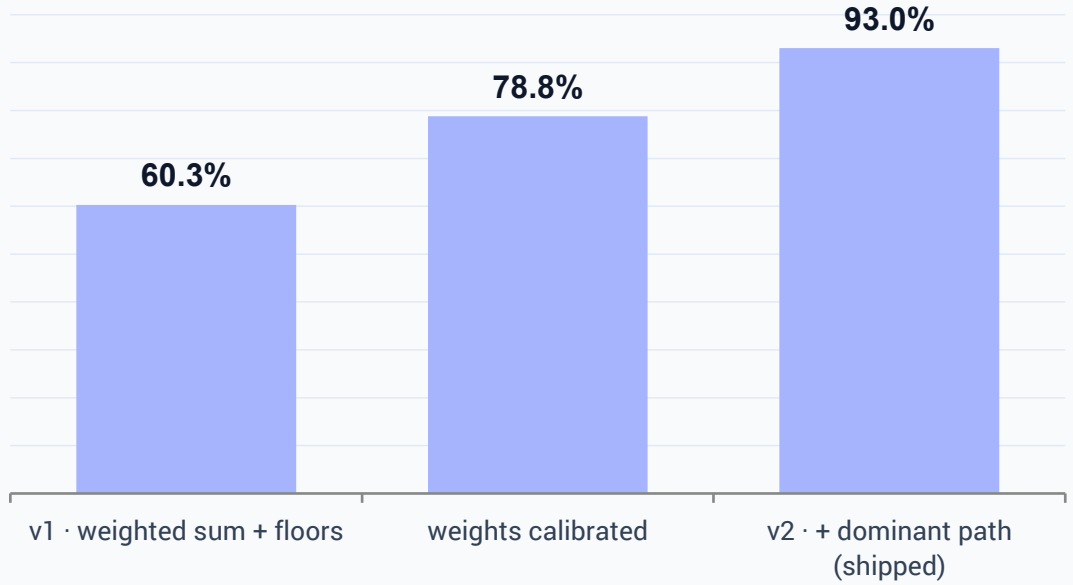
A weighted average measures agreement; a dominant path and floors stop one strong red flag being averaged away.

Component weights (normalised over what's available)



1. **Weighted average** Each component scores 0–1; only components with evidence count.
2. **Dominant path** fused = max(average, strongest signal × its confidence) — plus 0.1 if another signal agrees.
3. **Floors** Confident face mismatch · exact duplicate identity · tampering HIGH. One floor → at least MEDIUM; two → HIGH.
4. **Tiers, advisory only** MEDIUM ≥ 0.20 · HIGH ≥ 0.60. The score never rejects on its own.

Held-out tier accuracy · 400 labelled scenarios



96.7% of 300 fraud cases built from real forged documents, real faces and names are rated MEDIUM or HIGH (target ≥ 85%); one clean case in 300 is rated HIGH. On the 400 synthetic scenarios above, v2 scores 93%.

Statement patterns and fraud triage

Patterns read the bank statement row by row; triage looks across all applications for rings.

#14 FRAUD PATTERNS

PDF text layer
or OCR words

→

Tilt-corrected rows
date + amounts

→

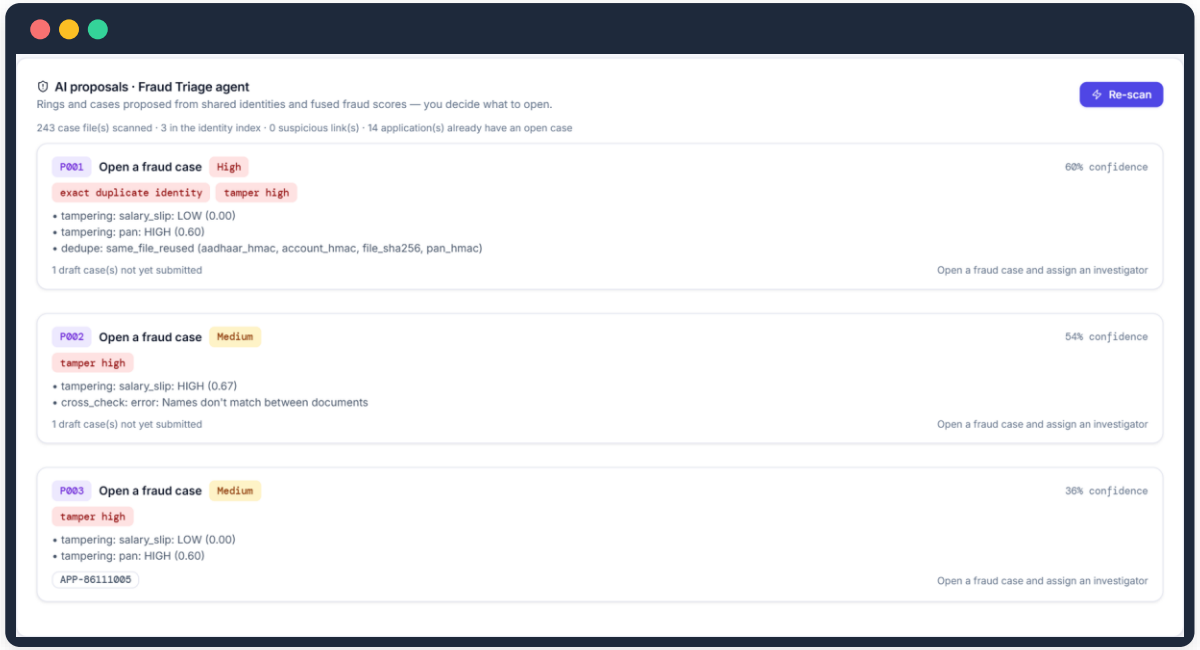
Reconcile
debit / credit by balance

FIVE PATTERNS

- **Cash cycling** money in and straight back out
- **Sudden deposit** a lump sum just before applying
- **Window dressing** balance pumped near statement end
- **Income mismatch** salary credits ≠ payslip net
- **GST inflation** credits far below declared turnover

F1 0.92

Pattern detection · 600 real accounts
statement rows recovered on Indian layouts: 4% → 95%

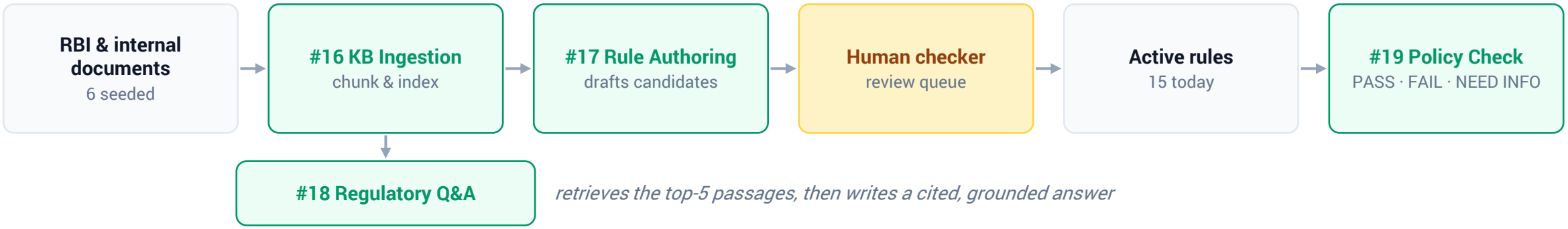


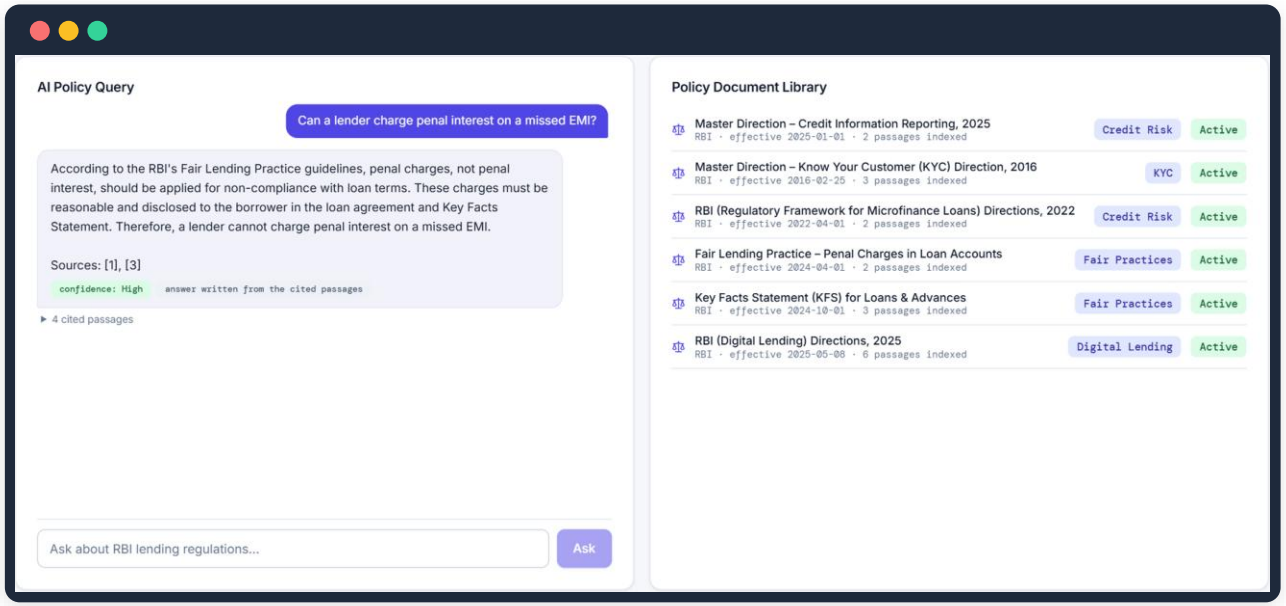
#15 FRAUD TRIAGE

- Links applications that share one face under several PANs, one bank account, one PAN with two faces, or one file
- Proposes cases only — a person decides what to open
- 80 planted rings among 1,000 real faces: every proposal was a real ring (71 false rings before the stricter 1:N face threshold)

Compliance: RBI rules as data, answers with citations

Regulations are indexed once; rules are drafted by an agent and approved by a person; every case is checked against them.





92.7% — **KB Ingestion**
right document in top 5 · 1,000 questions

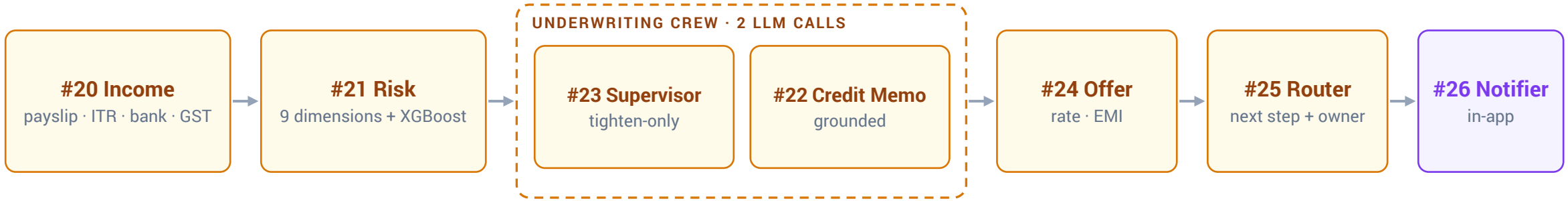
83.7% — **Rule Authoring**
77 of 92 labelled obligations drafted as rules

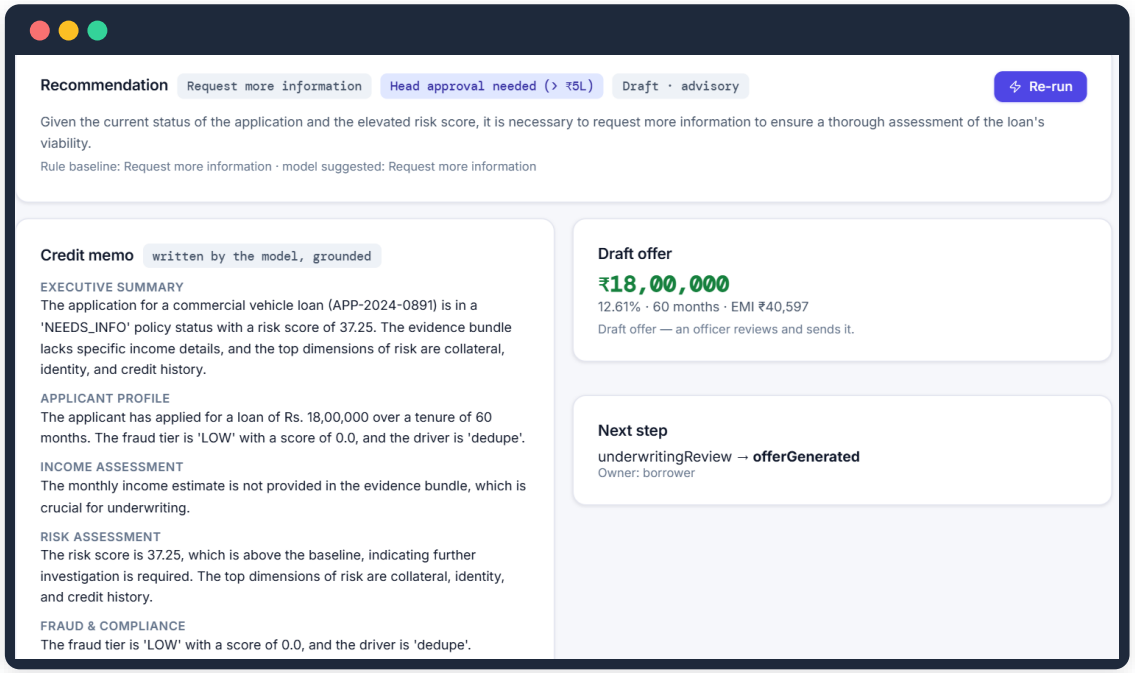
79.4% — **Regulatory Q&A**
an answering passage in the top 10

100% — **Policy Check**
116,000 verdicts equal an independent check

Underwriting: from income to a drafted offer

One click runs the chain; the only model calls are the two agents inside the crew. About 20 seconds.





F1 1.00 — **Income Analysis**
inflated income flagged on 400 real account histories

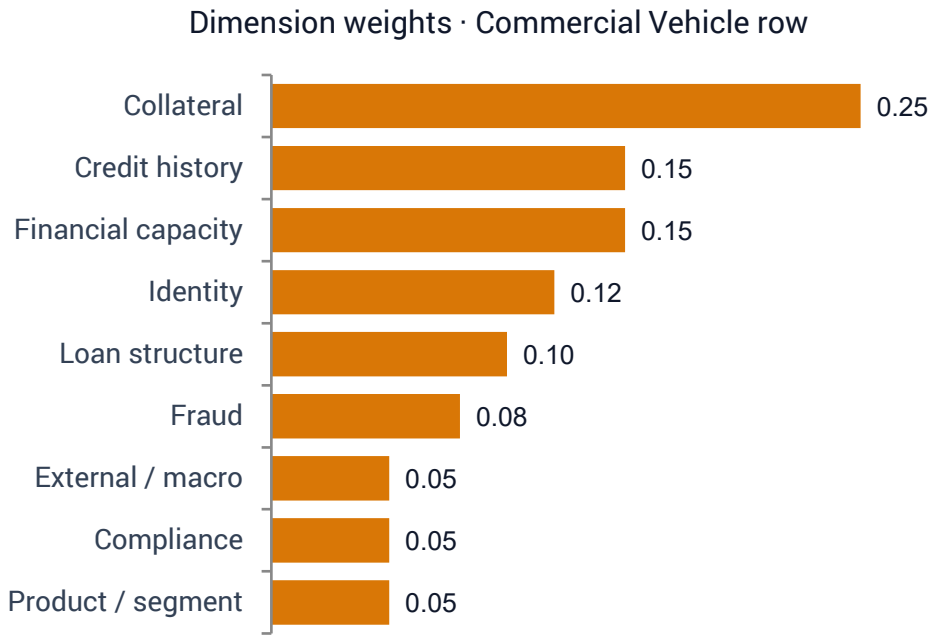
100% — **Credit Memo**
120 memos on real loans cite only real figures

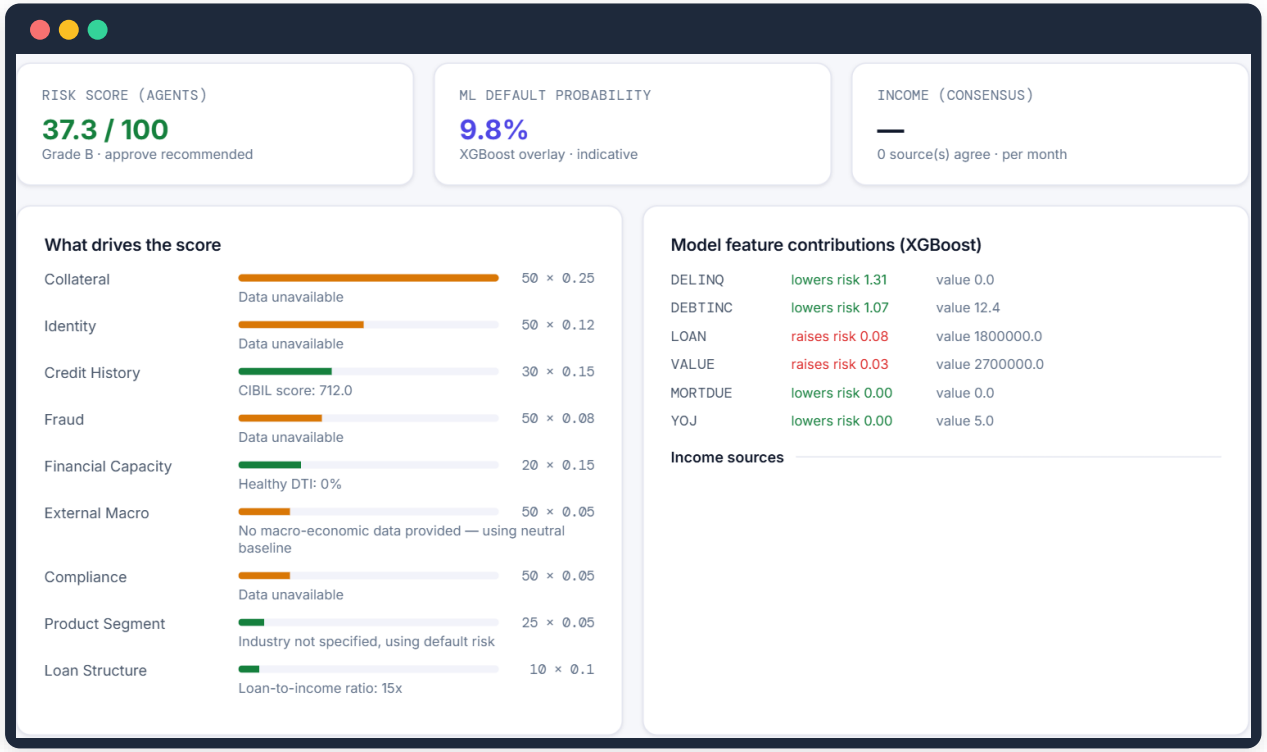
0.72× — **Supervisor**
defaults among loans it would approve vs all loans

100% — **Offer · Router**
EMI = formula (2,000 offers) · 147 status moves

Risk Scoring: nine dimensions plus a machine-learning check

A 12 × 9 weight matrix (one row per product) gives the grade; a trained default model adjusts it within ±25 points.





Application › Risk Analysis: agent score, ML default probability and what drives each.

66% → 83%
of a trained model's skill on 5,000 real two-wheeler loans (target ≥ 75%)

0 of 750
checks where a worse input lowered the risk score

Operations: notify, keep the record, watch the process

Three back-office agents. None of them can send anything outside the app.

#26 Notifier

Drafts notification rows from templates for borrower, officer and head.

1,000 notifications, all 16 events: 100% correct · 0 outbound attempts

#27 Audit / Storage

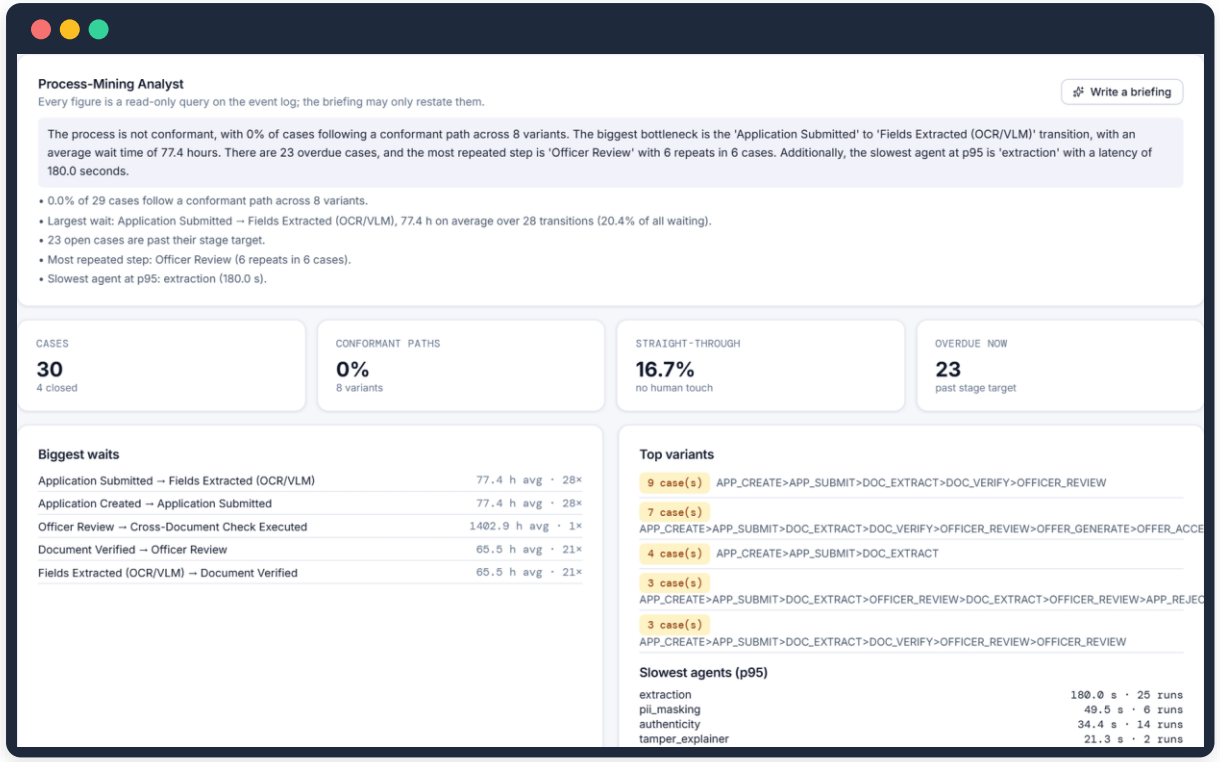
Persists files and SQL rows; officers see a masked copy.

120 real files intact · no PAN or Aadhaar on the officer sheet

#28 Process-Mining Analyst

Conformance, variants, waits and slow agents from pm.* views; optional grounded briefing.

38 of 38 figures equal pm4py on a real bank's event log (BPI 2017)



Agent Activity › Process Mining — seeded demo process data: 0% conformant paths, 23 cases overdue.

03

Workflows

How a loan moves through its lifecycle, who acts at each step, and what a real run looks like.



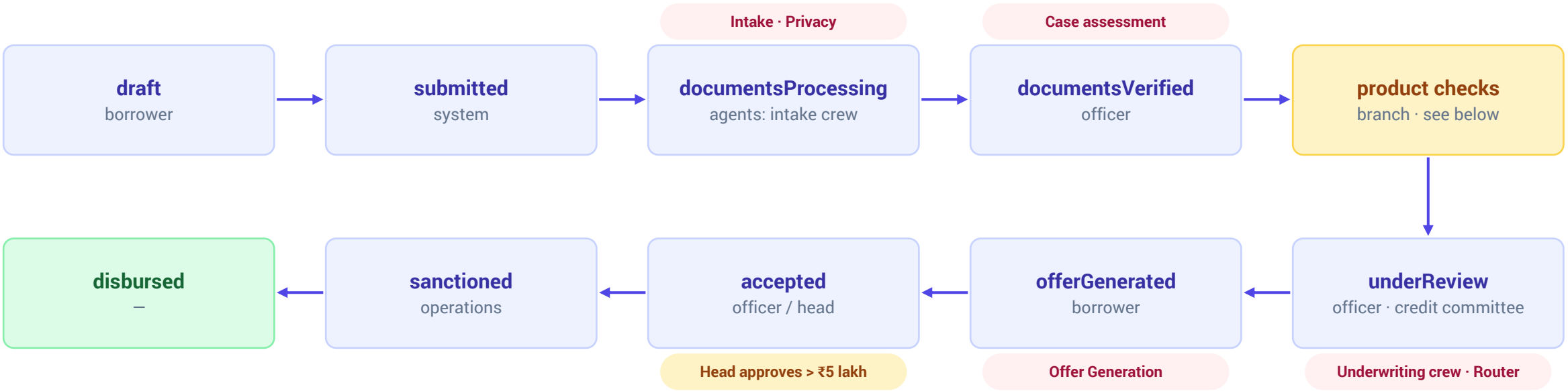
IN THIS SECTION

Loan lifecycle | End-to-end journey | A real run

WORKFLOWS

The loan lifecycle

One path for every product plus its own branch. The Workflow Router only recommends; a person moves the status.



PRODUCT BRANCHES (between documentsVerified and review)

Gold valuation scheduled → valued

Tractor · Microfinance field verification

Loan Against Property property valuation → legal

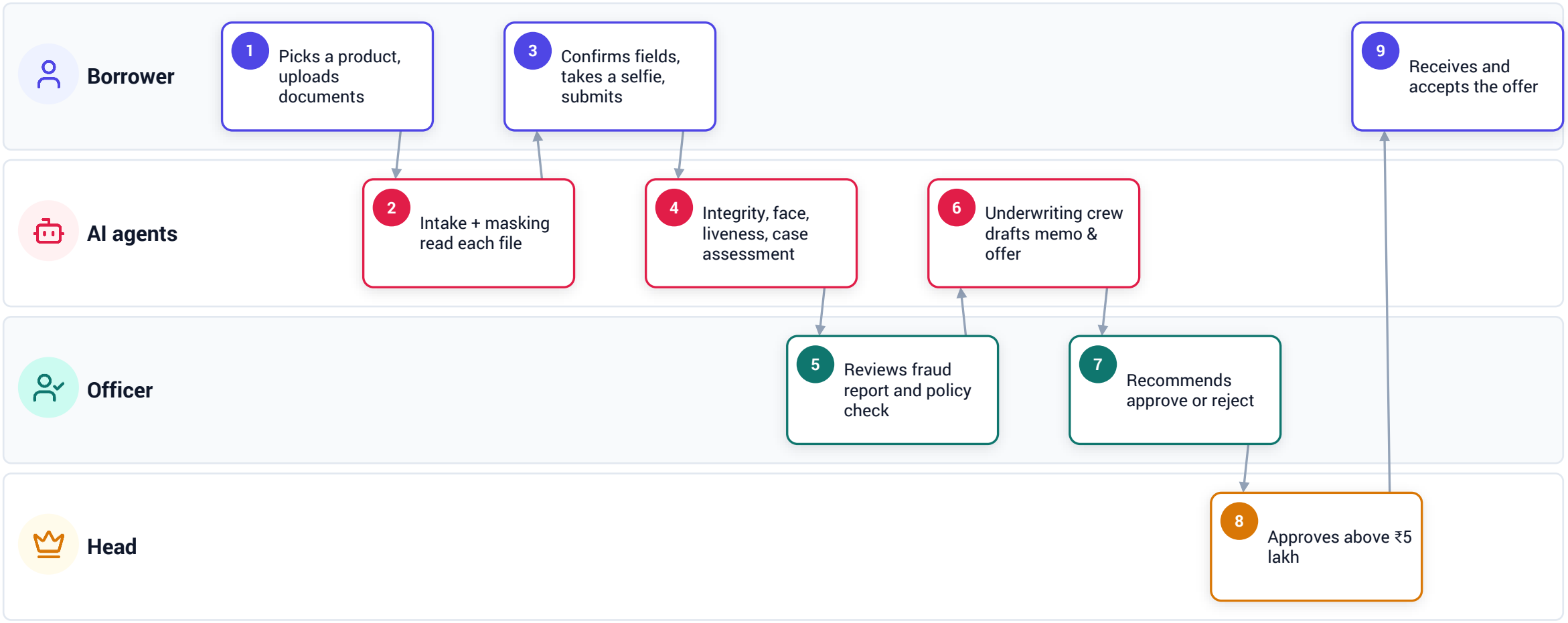
Home 6 checks: property, legal, income, docs, technical

MSME · Working capital bank & GST analysis → committee

Used vehicle vehicle inspection

End to end: who does what

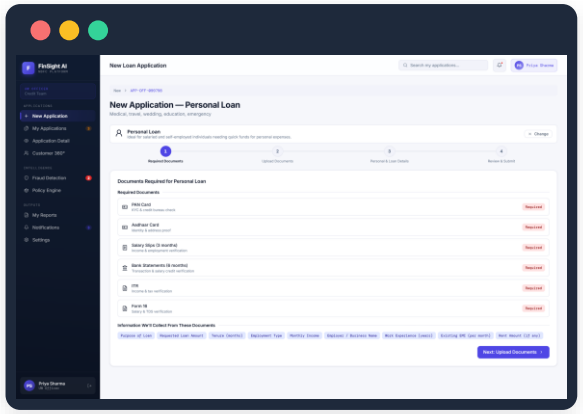
Agents work between the human steps; every agent output is evidence for the next person.



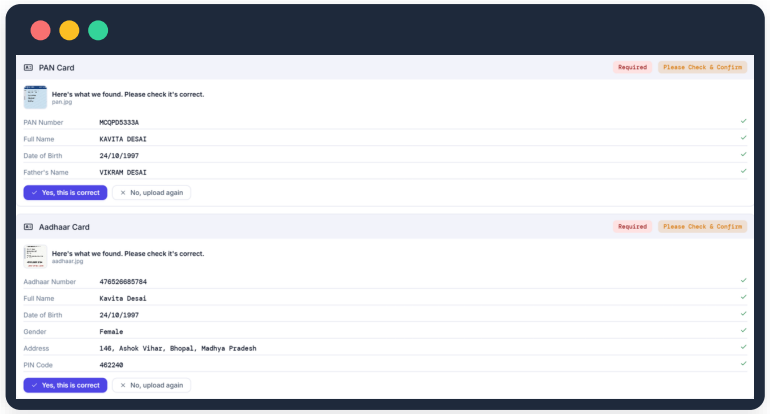
A real run through the wizard

Captured from the app with a synthetic applicant (generated identity — no real personal data).

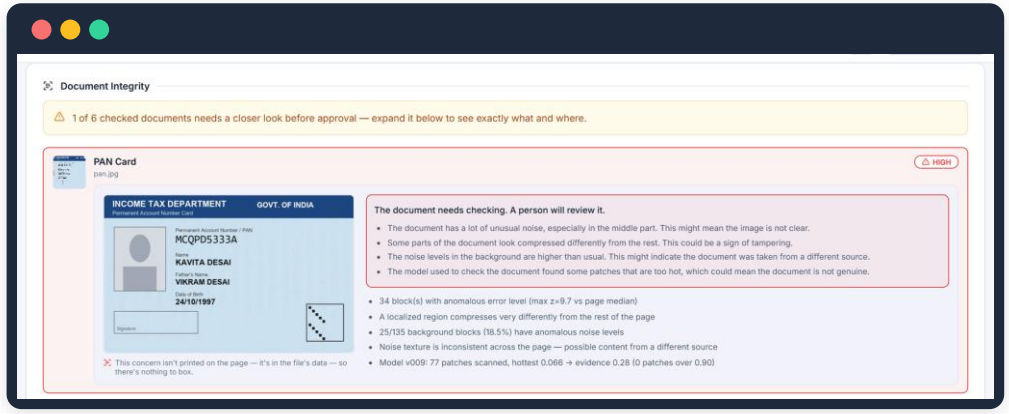
1 Product checklist



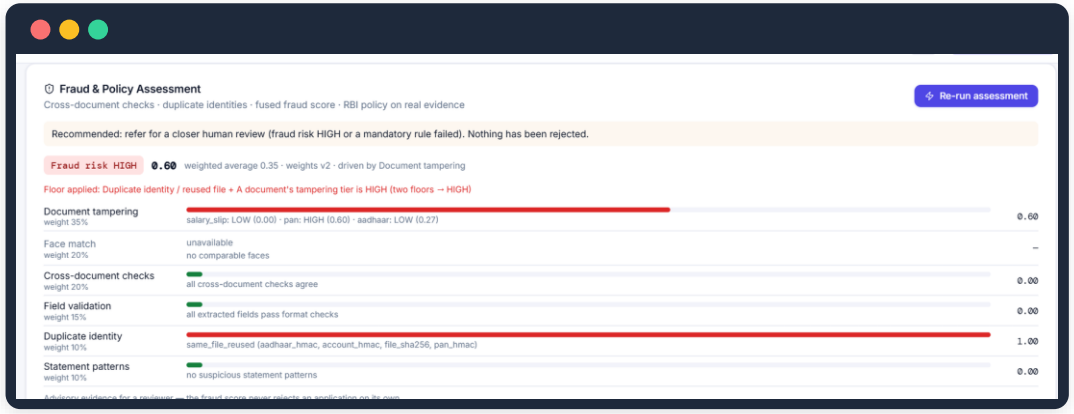
2 Fields read by the intake crew



3 Integrity: plain-language findings



4 Fraud & policy assessment



04

Trust & control

One gateway to every model, guardrails that are tested, and a trace of every step in SQL Server.



IN THIS SECTION

LLM gateway | Guardrails | Tracing | Data model

TRUST & CONTROL

One gateway for every model call

The Head Manager picks the model per agent, per lane or by default. The gateway enforces privacy and queues the calls.

Agent asks for a model
gateway.chat(agent_id, …)

↓

1 · Resolve the model
agent override → vision / text lane → default · snapshot per run

↓

2 · Privacy guard
FORBID_HOSTED switch · Head acknowledgement · pseudonymised text for hosted

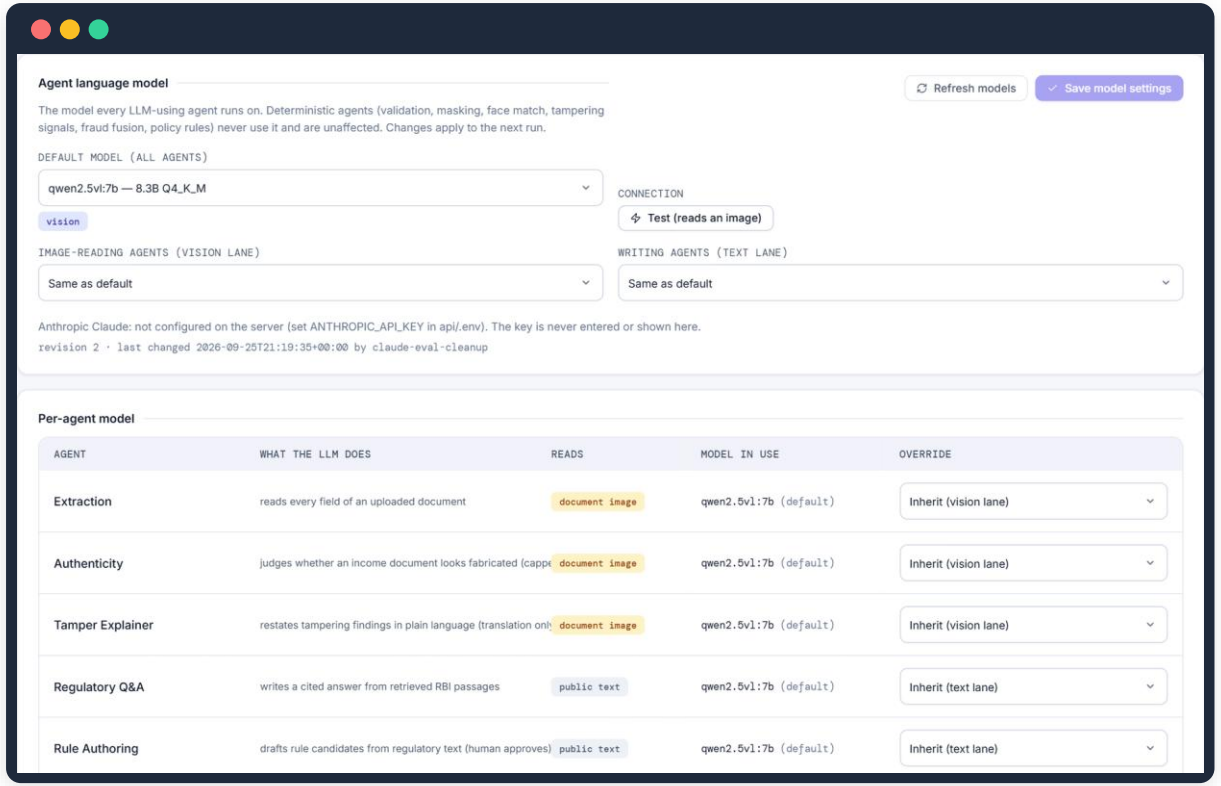
↓

3 · LLM gate
one slot per provider · priority queue · 180 s max wait

↓

Ollama (local)
127.0.0.1 · pooled · images fitted to context

Anthropic Claude
optional · not configured today



System Settings › AI Models (Head only): default, lanes and a per-agent override table.

TRUST & CONTROL

Guardrails — each one pinned by a test

The promises the design makes, and how each was verified.

LLM never sets a score

Scores, tiers and verdicts come from tools only.

tier unchanged 60 / 60

Grounding gate

Every number ≥ 10 in model text must be in the evidence.

ungrounded text dropped

Tighten-only supervisor

May make the next step stricter, never looser.

1 loosening attempt blocked

Nothing auto-rejects

Advisory routes only; above ₹5 lakh needs the Head.

router guard test

No side-effect tools

No agent can approve, reject, file, accept or send.

checked on all 29

LLM budget per path

/extract ≤ 1 call; masking, face and assessment 0.

call-count test

Authenticity capped

The fabrication signal is capped at 0.60, never alone.

cap held on all 418

Hosted models gated

Head acknowledgement, kill switch, pseudonymised text.

settings + gateway tests

Identity keys hashed

Dedupe stores HMAC digests, never raw PAN / Aadhaar.

holds

Contracts unchanged

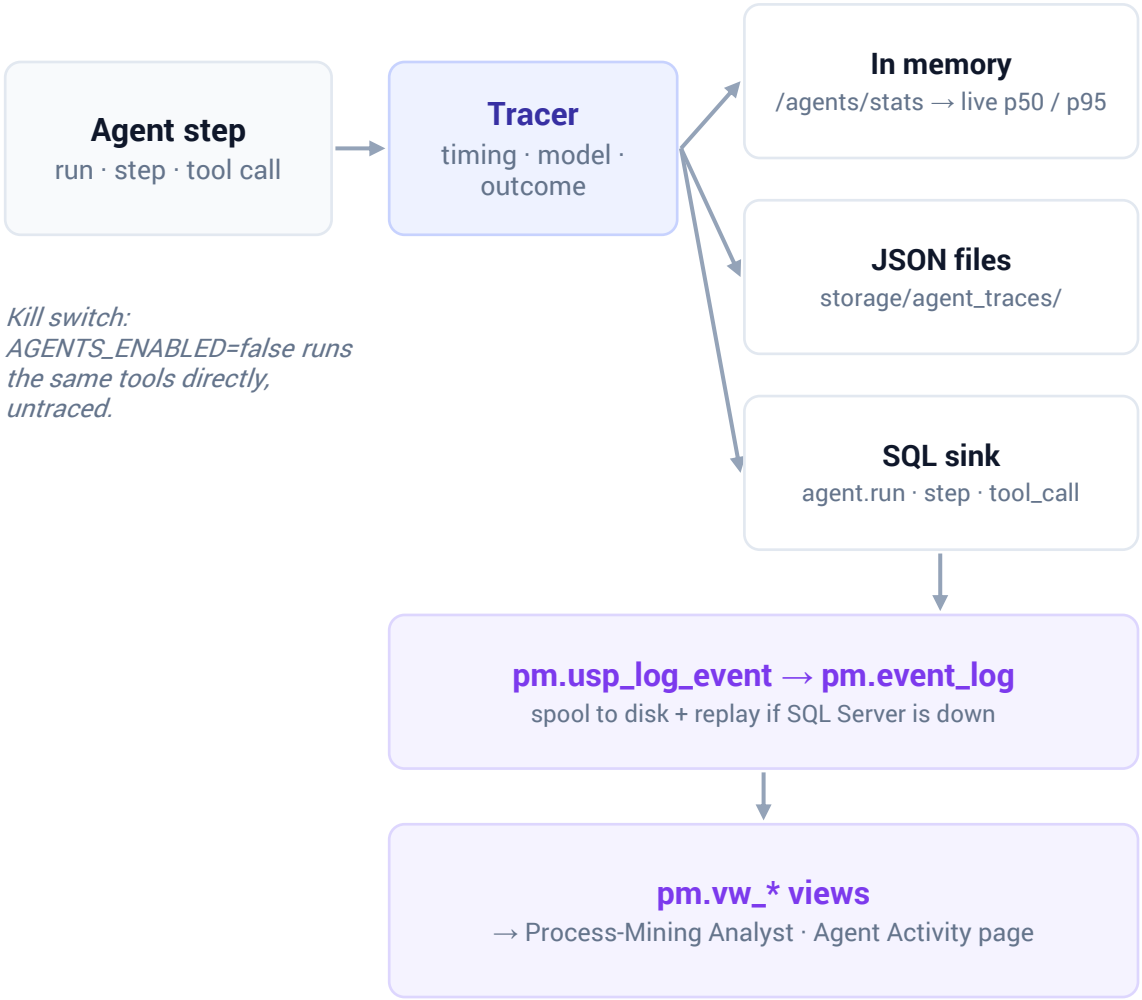
Pre-agent endpoints answer exactly as before.

24 / 24 contract tests

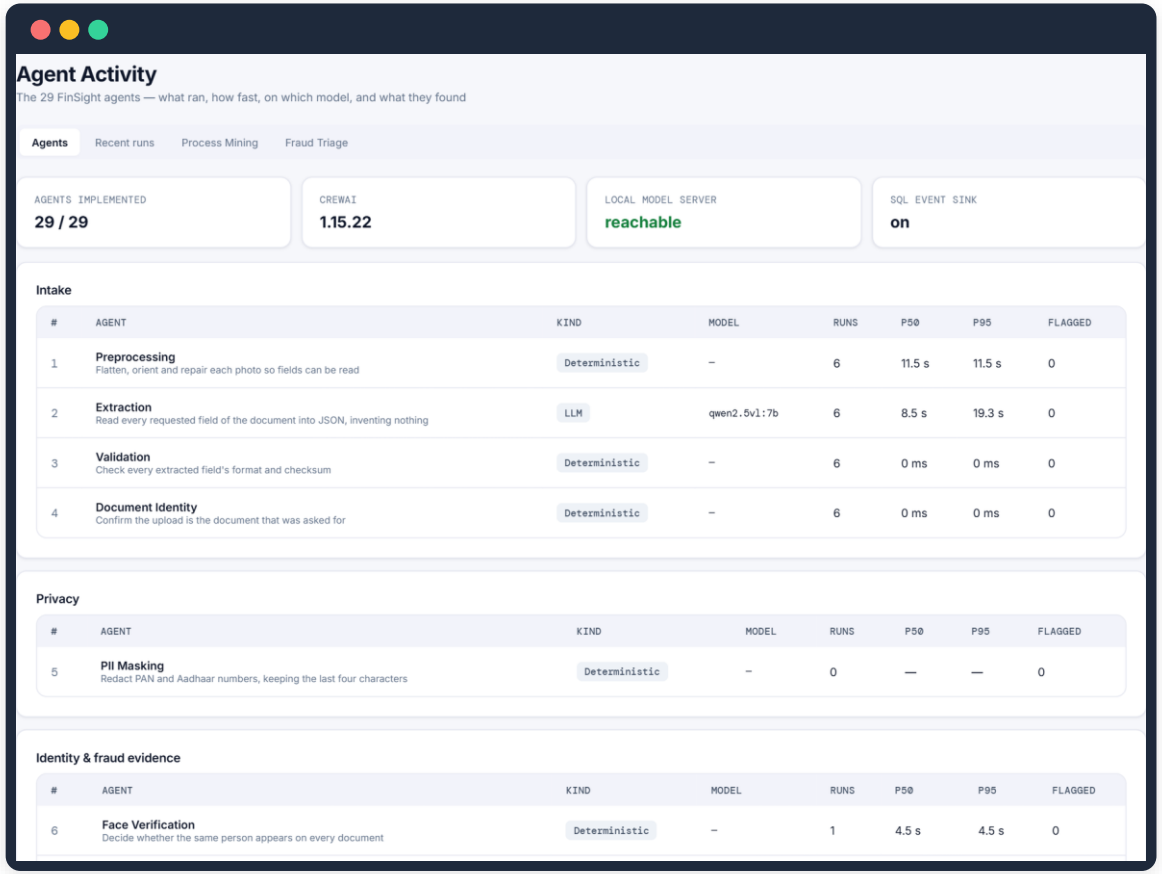
TRUST & CONTROL

Every step is traced — and becomes process-mining data

Runs and steps go to memory, JSON files and SQL Server. A database outage never fails a request.



Kill switch: AGENTS_ENABLED=false runs the same tools directly, untraced.



Agent Activity › Agents: each agent's kind, model, runs, p50 / p95 and flags.

TRUST & CONTROL

The database: SQL Server 'FinSight'

9 schemas · 82 tables · 17 views. Migrations live in docs/mssql/ — 08-agentic-layer.sql adds the agent tables.

core

9 tables

customers · loan_applications · uploaded_documents · document_extraction_results · loan_offers

fraud

9 tables

fraud_assessments · fraud_cases · fraud_rings · identity_keys · fraud_weight_sets

comp

5 tables

compliance_documents · document_chunks · rules · rule_candidates · check_results

risk

3 tables

risk_assessments · pricing_models · risk_thresholds

agent

4 tables

run · step · tool_call · llm_setting_audit

pm

20 tables

event_log · activity · case_metrics · case_variant · conformance_result · deviation

ops

8 tables

users · notifications · notification_templates · loan_products

audit

3 tables

audit_log · audit_log_details · business_events

ref

21 tables

application_status · lifecycle_transition · document_type · fraud_severity …

05

Evaluation

Tested twice: when each agent was built, and again on real and public data — against pass marks written down before the run.



IN THIS SECTION

Method | Scorecard | What real data changed | Finding real forgeries | Latency | Limits & next steps

EVALUATION

How the agents were tested: twice, with the target fixed first

Round 1 as each agent was built, on synthetic data. Round 2 on real and public data — the scores in this deck.

Fix the target
targets_real.yaml

Split the data
30% dev · 70% test

Run the app's own path
what an upload triggers

Fix on dev only
13 agents changed

Score test once
95% confidence interval

RULES OF THE TEST

- **Strong = target met**
- **Acceptable = within 10%**
- **Weak = below**
- **every model call on this PC**
- **no real PAN or Aadhaar numbers**
- **aggregate results only**

REAL AND PUBLIC DATA

Forged documents
Find it again!: 988 real receipts, 163 forged by people. SIDTD: ID images with replaced or repainted fields.

Photos and scans
MIDV-500 phone frames of ID cards; SROIE, FUNSD and RVL-CDIP real scanned pages.

Faces and liveness
AgeDB-30, CALFW, CPLFW and LFW pair protocols; NUAA and CelebA-Spoof printed-photo attacks.

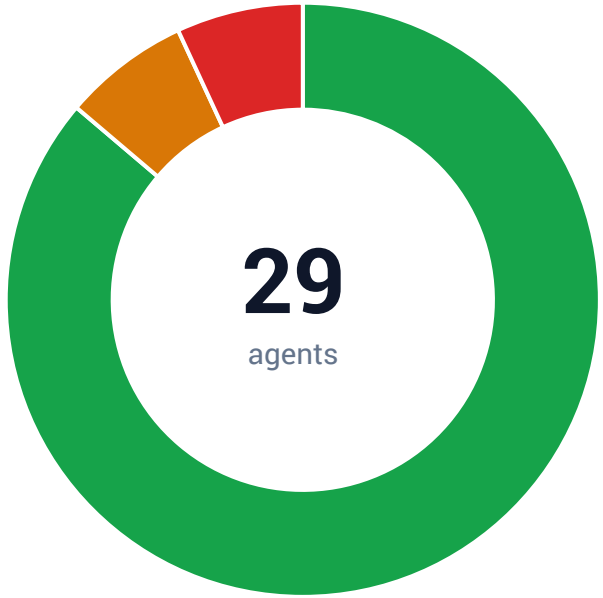
Loans and accounts
233,000 Indian two-wheeler loans with real defaults (L&T); 4,500 real bank accounts (Berka).

Regulations and process
ObliQA: 40 real regulations with gold answers. BPI 2017: a real bank's loan event log.

Indian IDs and statements
Labelled synthetic PAN / Aadhaar cards and statements in real bank layouts — no lawful public set of real ones exists.

EVALUATION

Scorecard on real data: 25 of 29 agents meet their target



25	Strong — target met
2	Acceptable — within 10%
2	Weak — below

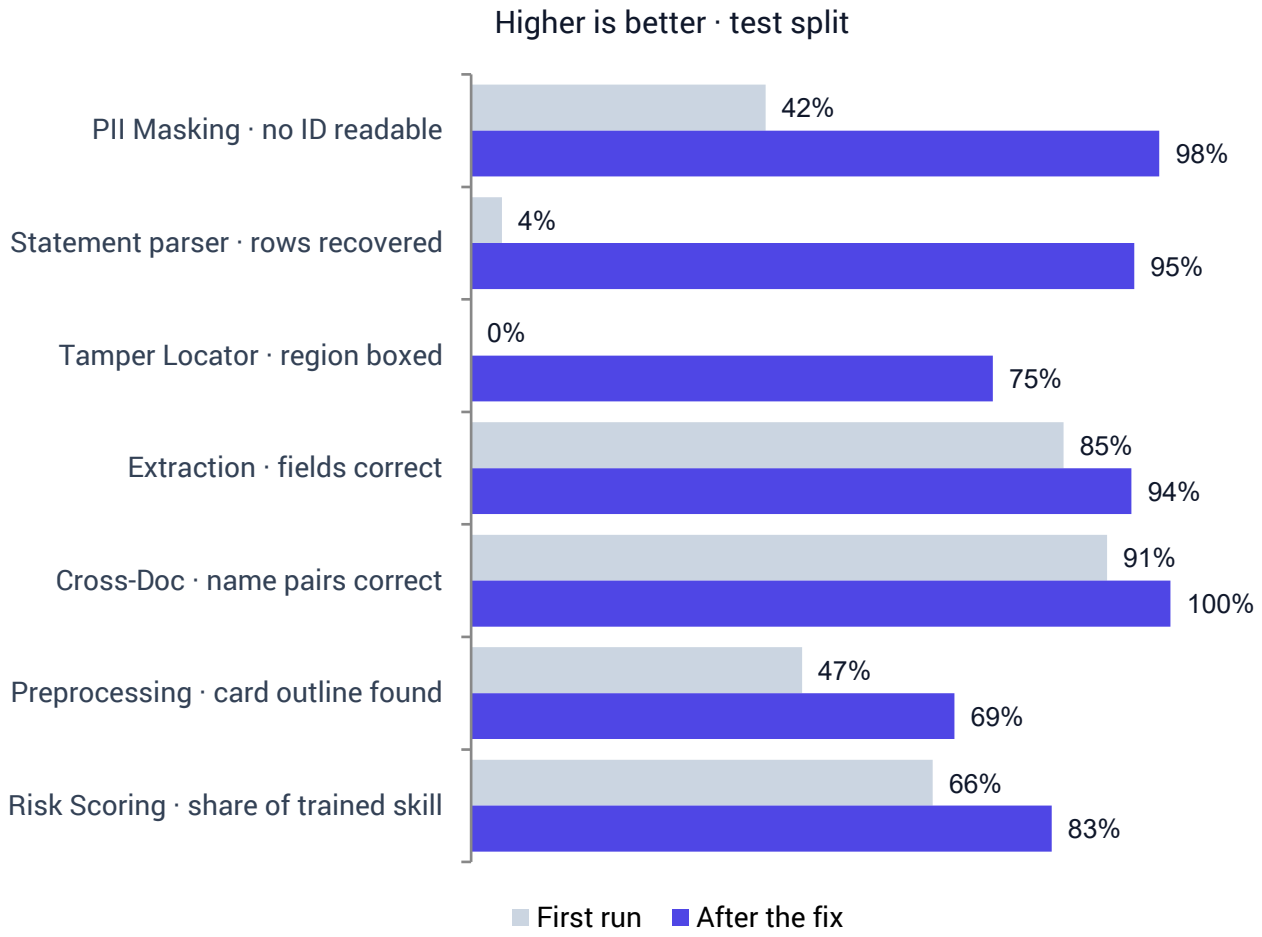
Held-out test splits · 95% intervals in report 08

1 Preprocessing 69.1% target ≥ 80%	2 Extraction 94.4% target ≥ 90%	3 Validation 100% target 100%	4 Document Identity 94.4% target ≥ 90%	5 PII Masking 98.4% target 100%
6 Face Verification 95.9% target ≥ 90%	7 Tampering Forensics 88.9% target ≥ 85%	8 Authenticity 11.2% target ≥ 70%	9 Tamper Explainer 98.3% target ≥ 95%	10 Tamper Locator 74.6% target ≥ 80%
11 Cross-Doc Verification 100% target ≥ 95%	12 Dedupe 96.3% target ≥ 95%	13 Fraud Fusion 96.7% target ≥ 85%	14 Fraud Patterns F1 0.92 target ≥ 0.85	15 Fraud Triage 100% target ≥ 80%
16 KB Ingestion 92.7% target ≥ 85%	17 Rule Authoring 83.7% target ≥ 70%	18 Regulatory Q&A 79.4% target ≥ 70%	19 Policy Check 100% target 100%	20 Income Analysis F1 1.00 target ≥ 0.90
21 Risk Scoring skill 0.83 target ≥ 0.75	22 Credit Memo 100% target ≥ 95%	23 UW Supervisor 0.72× target ≤ 0.85×	24 Offer Generation 100% target 100%	25 Workflow Router 100% target 100%
26 Notifier 100% target 100%	27 Audit / Storage 100% target 100%	28 Process Mining 100% target 100%	29 Liveness ACER 0.4% target ≤ 10%	

EVALUATION

Real data found weaknesses — 13 agents were fixed

Each fix was developed on the development split only; the test split was scored once afterwards. First run → after the fix.



LOWER IS BETTER

Dedupe new applicants given a false 'same face' alert	**41.8% → 0.4%**
Tampering Forensics genuine documents wrongly rated HIGH	**17.0% → 0.8%**
Document Identity right documents refused	**11.6% → 1.6%**
Preprocessing scans made harder to read by the cleanup	**30.7% → 6.0%**
Fraud Triage false fraud rings proposed	**71 → 0**

Also fixed: the Supervisor could never recommend 'approve' — loan-terms rules were checked before any offer existed.

EVALUATION

Finding real forgeries: from 17% to 89%

The pixel signals were built on synthetic pages. Real forged documents exposed two problems, fixed one after the other.

1

As first configured

84% of genuine documents flagged

Signals scaled on synthetic pages saturate on real scans; forged documents ranked below genuine ones (AUC 0.36).

2

Calibrated on genuine documents

False alarms gone — but 17% of forgeries found

A pixel signal counts only above what 95% of genuine documents reach; ELA and noise need a second signal.

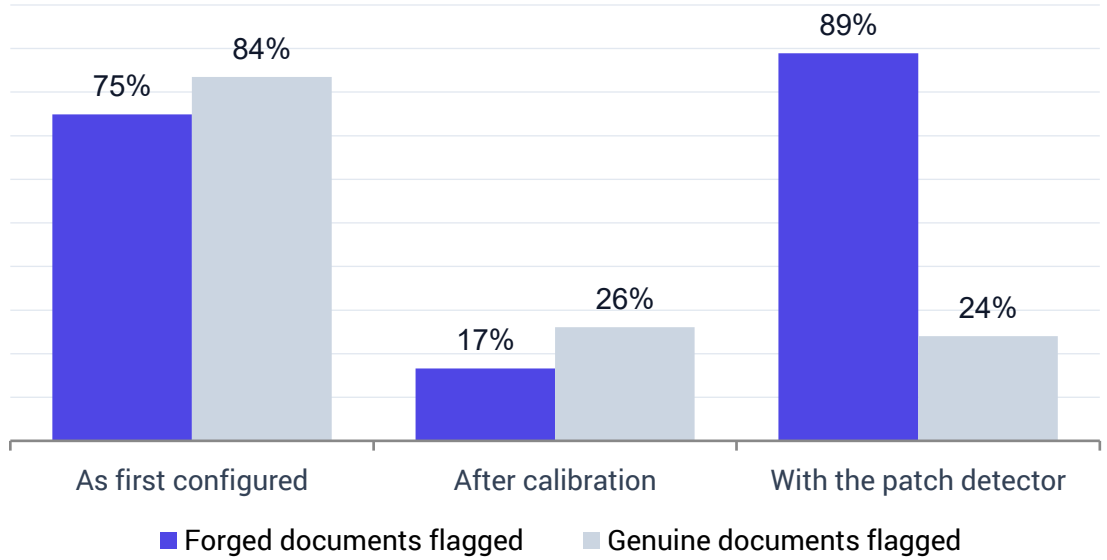
3

A detector trained on real forgeries

89% of forgeries found

Two small patch models per document family, each calibrated on documents it never saw. Evidence only when both agree; alone it gives MEDIUM, never HIGH. It names the region, so the locator can draw it.

Test split · 235 forged and 383 genuine documents



54% of receipts edited by people (Paint, GIMP)

95% of ID images forged by script

0.5% of genuine ID phone photos flagged

It has never seen a forged Indian document — none exists to test on — so its tier stays advice to a reviewer.

EVALUATION

Build-time testing found — and fixed — 17 defects

Round 1. The eight that mattered most; each was caught by an agent's evaluation or the latency benchmark, not by users.

PDF handling

Two threads inside PDFium at once crashed the whole server — with no error logged.

Fix One process-wide PDFium lock.

PII Masking

Sideways cards split the wrong way; one real PAN photo leaked in full.

Fix Upright rotation first: 0 leaks.

Document Identity

A payslip showing a PAN passed as a PAN — 54% of wrong uploads caught.

Fix Also reads the page's own words: 100%.

LLM gateway

'localhost' tried IPv6 first on Windows: +2 s on every model call.

Fix 127.0.0.1 and one pooled connection.

Risk Scoring

ML features built unlike training: held-out AUC 0.669.

Fix Training feature builder reused: 0.771.

Policy Check

Face rule used another model's 0.62 scale: every genuine match failed.

Fix SFace threshold 0.363.

Structured LLM calls

The model echoed the schema: 92% of written Q&A answers discarded.

Fix Schema-enforced JSON output.

Cross-Doc Verification

A shared surname alone counted as a name match — family members looked like one person.

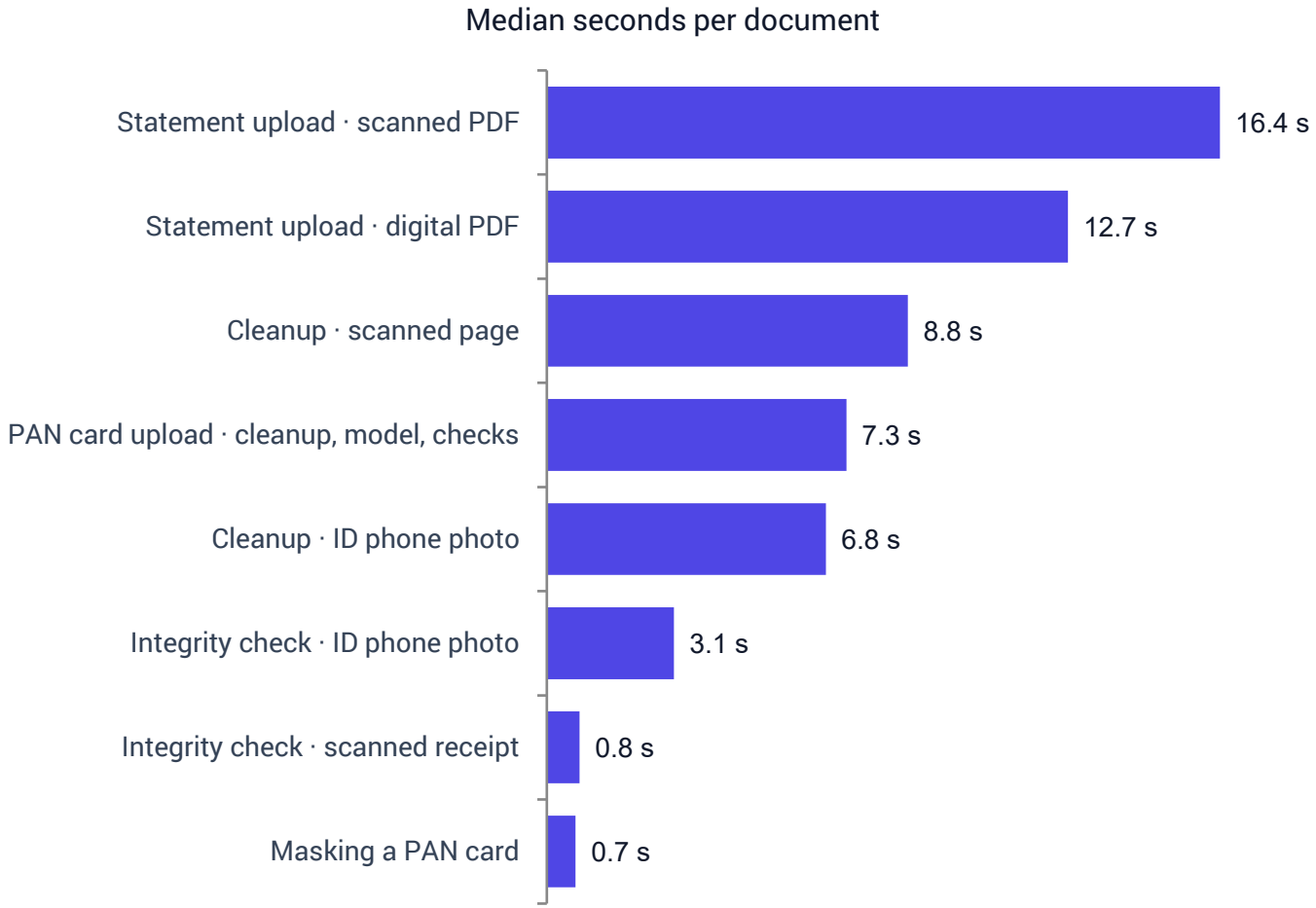
Fix Fixed in the browser and Python logic.

+ 9 more: card rules applied to full pages · images overflowing the model context · regulation titles ignored by grounding · unreadable photo rows treated as real · SQL inserts refused on tables with triggers · a parser error result · review statuses in the router · a stale Agent Activity tab · PDFs rasterised needlessly

EVALUATION

Latency on one laptop

Median seconds per document on an idle machine, local model already loaded (RTX 4060, 8 GB).



THREE FIXES FOUND WHILE BENCHMARKING

2 s on every model call
Windows tried IPv6 for 'localhost' first — now 127.0.0.1, pooled.

ID photos 19% slower
A second OCR pass on ID cards — now reuses preprocessing's words.

Statement photos ~23 s
Their rows are now parsed in the background after the upload.

Agents did not slow the app. Against the pre-agent build the two model uploads got 9–11% faster; every other endpoint added at most 56 ms.

EVALUATION

Known limits and open decisions

LIMIT	NEXT STEP
Authenticity · Weak 11% The local 7B vision model answers 'edited' for genuine and forged areas alike	Leave local edits to the patch detector
Card outline · Weak 69% A pale card on a pale surface, or a cluttered background, is missed	Ask for a plain dark background; train on real photos
Tamper Locator · 74.6% A document flagged for another reason (an editor tag) gets no box	Regions from more signals to reach 80%
PII Masking · 98.4% A card whose number can't be located is withheld (about 1 PAN in 11)	Redact those by hand; add a field detector
No forged Indian documents Tampering was tested on real receipts and European ID images	Collect officers' review verdicts as labels
RBI circulars Compliance was tested on 40 ADGM regulations; rbi.org.in blocks scripts	Download six circulars by hand, re-run

OPEN DECISIONS

1. **What MEDIUM should mean**
   About half of clean test cases get a MEDIUM fraud tier (24% with real scans): a genuine file with an editor tag is flagged.

2. **Verify the PAN with the issuer**
   83% of PAN numbers are read correctly, and a wrong read still looks like a valid PAN.

3. **Watch the first live flags**
   The detector's floors were measured on other kinds of documents than Indian ones.

GETTING STARTED

Run it yourself

Five steps on one Windows machine. Ports: frontend 5173, API 8000, Ollama 11434.

1 Database

```
run docs/mssql/*.sql
on (localdb)\MSSQLLocalDB
```

2 Local model

```
ollama pull qwen2.5vl:7b
```

3 API :8000

```
cd api
python -m venv .venv-agents
.venv-agents\Scripts\pip install -r requirements-agents.txt
run.bat
```

4 Frontend :5173

```
npm install
npm run dev
```

5 Tests & evaluation

```
pytest tests
python -m eval.real.run_real --agent <id>
python -m eval.real.report_real
```

WHERE THINGS LIVE

api/app/agents/ registry, agent implementations, flows, tools, llm/gateway.py

api/eval/real/ real-data harness · targets_real.yaml · results_real/

docs/reports/08-…Real-Data-Evaluation.pdf the full per-agent report

github.com/ShreyasBile/FinSight source, reports and this deck

FinSight AI

Agents read, check and draft. People decide.

29	25	13	0
agents in 6 crews	meet their target on real data	agents improved by real-data testing	auto-rejections

Local model by default · every number grounded · every step traced · every decision human

Thank you · Questions?

github.com/ShreyasBile/FinSight